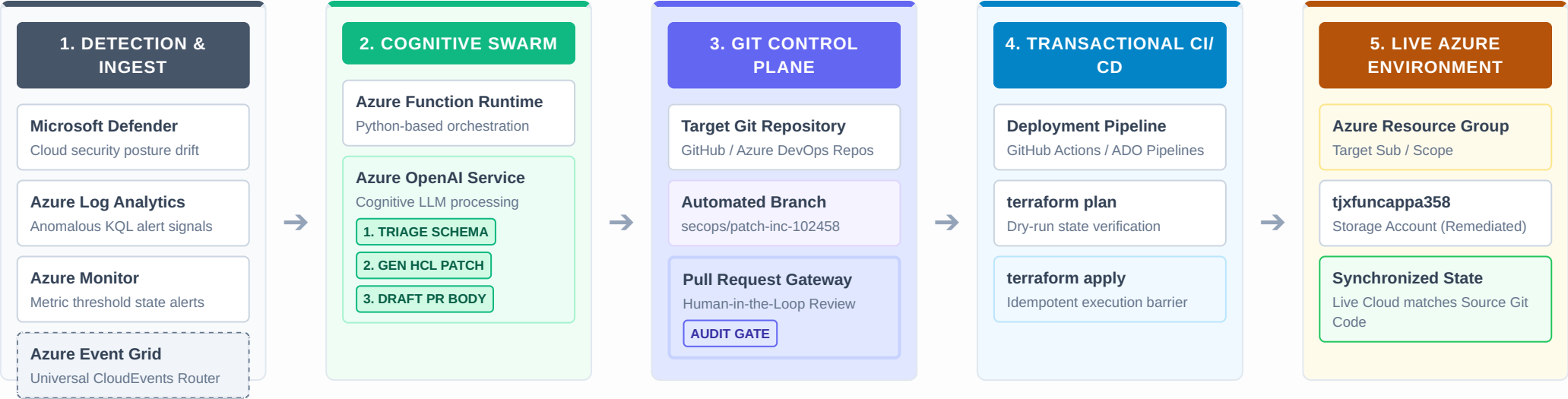


AzureOps Sentinel: Enterprise GitOps Blueprint

A Multi-Source Event-Driven Declarative Remediation Architecture for Terraform-Managed Public Cloud Infrastructures

1. End-to-End Event-Driven Functional Architecture Diagram



2. Deep Dive Lifecycle Mappings & System Specifications

Lifecycle Phase	Component & Technical Interface	Core Operational Mechanics	OpenAI Cognitive Footprint
DIAGNOSING	Azure Event Grid System Topic INGESTION LAYER monitoringService: "Defender"	Ingests polymorphic JSON structures from divergent telemetry planes. Standardizes communication via the CloudEvents v1.0 standard, shielding downstream code modules from Microsoft event schema modifications.	Footprint #1: Cognitive Parser Extracts nested variables (e.g., compromisedEntity) into a unified internal JSON definition without complex regex code.
REMEDIATING	Azure Function (Python) & GitHub API ORCHESTRATION LAYER	The function checkouts out the repository context, passes the affected .tf block to OpenAI, handles branch creation, commits the altered declarative state, and programmatically spawns a Pull Request.	Footprint #2: Generative HCL Patching Analyzes existing configurations against policy violations and modifies specific attributes safely

Lifecycle Phase	Component & Technical Interface	Core Operational Mechanics	OpenAI Cognitive Footprint
			(e.g., turning <code>public_access</code> off).
VERIFYING	CI/CD Engine & Terraform Core VALIDATION LAYER terraform plan -out=tfplan	Triggered dynamically upon PR creation. Runs transactional dry-runs against the active <code>.tfstate</code> storage backend to verify structural integrity, syntactic validity, and identify structural side effects before changes touch target subscriptions.	None. (Decoupled completely from LLM code generation to enforce strict deterministic safeguards and strict state file alignment).
REPORTING	Pull Request Interface & Git Ledger AUDIT PLANE	Consolidates automated analytical breakdowns directly within the standard engineering workflow. The merged branch generates an immutable, cryptographically signed cryptographic ledger record inside the version control database.	Footprint #3: PR Copywriting Generates comprehensive Markdown documentation detailing Business Impact Summaries, blast radiuses, and required compliance review metadata.